

Deep Learning

UW ECE 657A - Core Topic

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Lecture Outline

- 1 Recall: Do you Remember Linear Classifiers?
 - Linear Regression
 - Logistic Regression
 - Another Linear Classifier - Naive Bayes
 - Kernel Methods - Turning Linear into Non-linear
- 2 Artificial Neural Networks
 - Simple Neural Network Definition
- 3 History
- 4 How Simple Neural Networks Work
- 5 Neural Networks as Universal Approximators
- 6 Training Neural Networks - Backpropagation
- 7 References and Further Reading

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Linear Regression vs. Logistic Regression

- A simple type of **Generalized Linear Model**
- Linear regression learns a function to predict a continuous variable output of continuous or discrete input variables

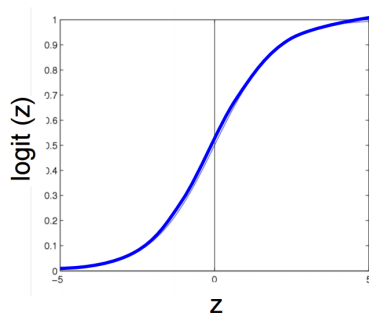
$$= b_0 + \sum (b_i X_i) + \epsilon$$

- Logistic regression predicts the probability of an outcome, the appropriate class for an input vector or the **odds** of one outcome being more likely than another.

Logistic Regression

Define probability of label being 1 by fitting a linear weight vector to the logistic (a.k.a sigmoid) function.

$$\begin{aligned}
 P(Y = 1|X) &= \frac{1}{1 + e^{-(w_0 + \sum_i w_i x_i)}} \\
 &= \frac{1}{1 + \exp(-(w_0 + \sum_i w_i x_i))}
 \end{aligned}$$

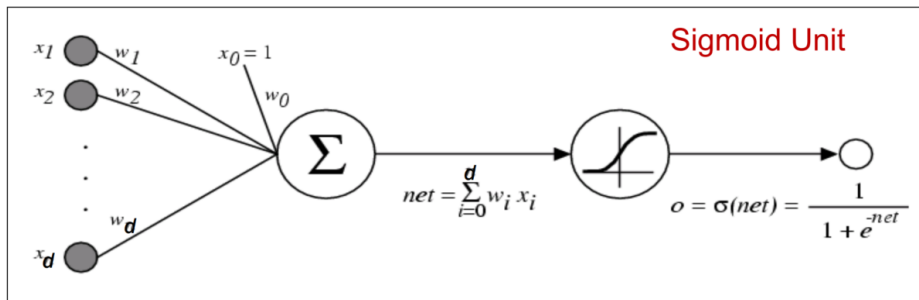


Advantage of this is you can turn the continuous $[-\infty, \infty]$ feature information into $[0, 1]$ and treat it like a probability.

bias: w_0 is called the bias, it basically adjusts the sigmoid curve to the left or right, biasing what the expected outcome is. The other weights w_i adjust the steepness of the curve in each

Logistic Regression as a Graphical Model

$$o(x) = \sigma(w^T x_i) = \sigma(w_0 + \sum_i w_i x_i) = \frac{1}{1 + \exp(-(w_0 + \sum_i w_i x_i))}$$



Properties of Logistic Regression

- Very simple yet powerful classification method
- Relatively easy to fit to data using gradient descent, many methods for doing this
- Learned parameters can be interpreted as computing the **log odds**, the logarithm of the ratio of successes to failures

$$\begin{aligned} LO &\sim \log \frac{p(Y = 1|X)}{p(Y = 0|X)} \\ &= \mathbf{w}^T \mathbf{x} \end{aligned}$$

- If x_0 : number of cigarettes per day
- x_1 : minutes of exercise
- $y = 1$: getting lung cancer
- Then if parameters learned at $\hat{\mathbf{w}} = (1.3, -1.1)$, it means each cigarette raises odds of cancer by factor of $e^{1.3}$

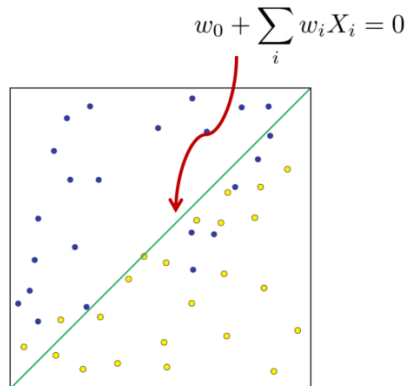
Logistic Regression Used as a Classifier

Logistic Regression can be used as a simple linear classifier.

- Compare probabilities of each class $P(Y = 0|X)$ and $P(Y = 1|X)$.
- Treat the halfway point on the sigmoid as the decision boundary.

$P(Y = 1|X) > 0.5$ classify X in class 1

$$w_0 + \sum_i w_i x_i = 0$$



Training Logistic Regression Model via Gradient Descent

- Can't easily perform Maximum Likelihood Estimation
- The negative log-likelihood of the logistic function is given by NLL and it's gradient by g

$$NLL(w) = \sum_{i=1}^N \log \left(1 + \exp(- (w_0 + \sum_i w_i x_i)) \right)$$
$$g = \frac{\partial}{\partial w} = \sum_i (\sigma(w^T x_i) - y_i) x_i$$

Then we can update the parameters iteratively

$$\theta_{k+1} = \theta_k - \eta_k g_k$$

where η_k is the learning rate or step size.

Other Linear Classifiers

- k Nearest Neighbours - label data points based on similarity to others using distance metric over defined neighbourhood.
- Naive Bayes - probabilistic model that assumes independence of input features. Uses Bayes rule to update a distribution over the inputs to minimize entropy of the outputs. Classification by threshold on probability.
- Support Vector Machines - learns a discriminant function and threshold to divide points into two classes.

More Powerful ML Classifiers

Naive Bayes is a probabilistic model that assumes independence of input features and uses Bayes rule to update a distribution over the inputs that minimizes entropy of the outputs. Classification is performed by choosing the more likely class or apply a threshold τ to determine if a class is valid.

Support Vector Machines (SVMs)

- SVM learns a discriminant function and threshold to divide points into two classes.
- Originally this was a line $w^T x + b$ where weights w and offset b are learned so that positive training points are above the line.
- More generally we can learn a nonlinear function using the **kernel trick**.

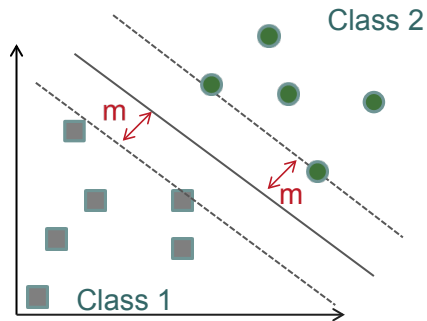
We rewrite the linear system as a dot product:

$$w^T x + b = b + \sum_{i=1}^m \alpha_i x^T x^{(i)}$$

where

- $x^{(i)}$ is a training example
- α is a vector of coefficients

Support Vector Machines



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We rewrite the linear system as a dot product:

$$w^T x + b = b + \sum_{i=1}^m \alpha_i x^T x^{(i)}$$
$$f(x) = b + \sum_i \alpha_i k(x, x^{(i)})$$

where

- $x^{(i)}$ is a training example
- α is a vector of coefficients
- $k(x, x^{(i)}) = \phi(x) \cdot \phi(x)^T$ is a kernel.

Pros and Cons of Kernel Methods

Pro:

- They map low dimensional linear space to higher dimensional nonlinear space.

Cons:

- Evaluation is expensive - linear in the weighted data points used $\alpha_i k(\mathbf{x}\mathbf{x}^{(i)})$
 - kernel SVMs avoid this problem because most of $\alpha_i = 0$, only the **support vectors** $\mathbf{x}^{(i)}$ are computed.
- Training is expensive - for large datasets
- Do not Generalize well - the kernel determines what kind of general patterns it can learn.
- Choosing the kernel is hard - too general or too specific to the problem, how to choose it?

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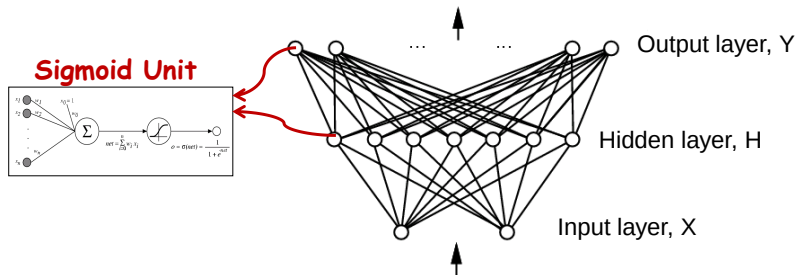
Neural Networks

- A Neural Network (NN) connects many nonlinear (classically logistic/sigmoid) units together into a network
- Central difference from the Perceptron: a layer of **hidden units**
- Also called: Multi-layer Perceptrons (MLP), Artificial Neural Networks (ANNs)

Neural Networks to learn $f : X \rightarrow Y$

- f can be a non-linear function
- X (vector of) continuous and/or discrete variables
- Y (vector of) continuous and/or discrete variables

Feedforward Neural networks - Represent f by network of non-linear (logistic/sigmoid/ReLU) units:



Basic Three Layer Neural Network

Input Layer

- vector data, each input collects **one feature**/dimension of the data and passes it on to the (first) hidden layer.

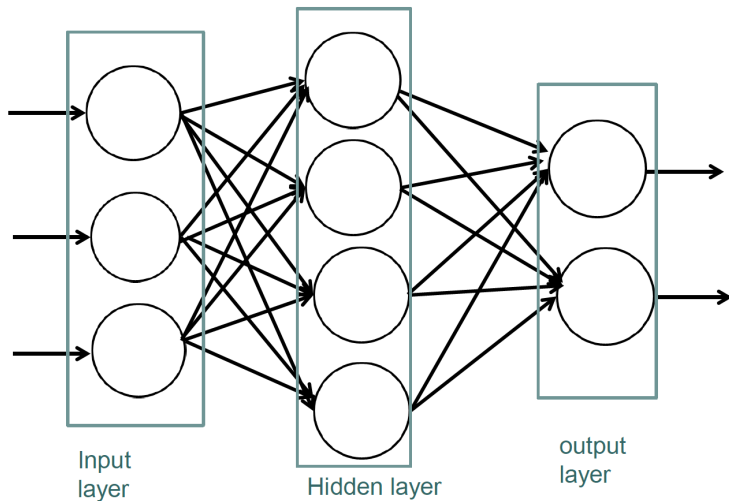
Hidden Layer

- Each hidden unit computes a weighted sum of all the units from the input layer (or any previous layer) and passes it through a **nonlinear activation function**.

Output Layer

- Each output unit computes a weighted sum of all the hidden units and passes it through a (possibly nonlinear) **threshold function**.

Three-layer Artificial Neural Network



Properties of Neural Networks

- Given a large enough layer of hidden units (or multiple layers) a NN can represent **any function**.
- One challenge is regularization: how many hidden units or layers to include in the network? Number of inputs and outputs are provided but internal structure can be arbitrary.
 - too few units, network will have too few parameters, may not be able to learn complex functions
 - too many units, network will be overparameterized and won't be forced to learn a generalizable model

Representation Learning: classic statistical machine learning is about learning functions to map input data to output. But Neural Networks, and especially Deep Learning, are more about learning **a representation** in order to perform classification or some other task.

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A Short History

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- 1958 Rosenblatt in 1958 introduced the **Perceptron**, a two layer network (one input layer and one output node with a bias in addition to the input features).
- 1969 Marvin Minsky: 1969. Perceptrons are 'just' linear, AI goes logical, beginning of "AI Winter"

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- 1990s SVMs! Kernels can do **anything!** (no, they can't)

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- 2012-present Google Cat Youtube, speech recognition, self driving cars, computer defeats world Go champion, ...

A “New” Age of Neural Networks

- Geoffrey Hinton (UofT, now Google Research), paper references
 - early work on back-propagation
 - Restricted Boltzman Machines - first deep architecture, idea for unsupervised pretraining of layers for RBMs was used for first successful use of deep networks.
 - Several applications of Deep Learning to image recognition, speech recognition,...
 - Regularization methods: dropout, data augmentation
- Yoshua Bengio (University of Montreal, MILA, ElementAI, ...)
 - ReLU improves efficiency
 - unsupervised representation learning of autoencoders
 - word2vec - learn general representation of words for classification, translation, etc.
 - General Adversarial Networks - an exciting new approach to training
- Yann LeCun (NYU, now Facebook AI Research)
 - Convolutional Neural Networks

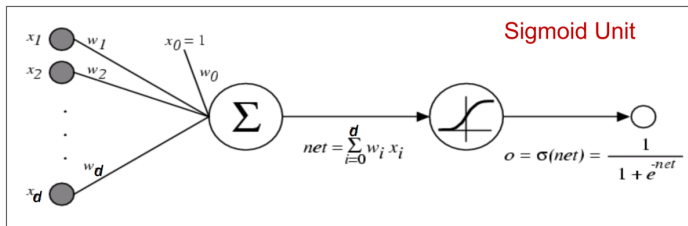
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Properties of Neural Networks

- The hidden layer will have a number of units (design parameter).
- The hidden layers allow extracting more complex features and facilitates solving complex problems.
- Connection between the units (neurons) of all the layers can be **forward**, backward or both.
- Units can be fully or partially connected.
- Different arrangements make different types or models of the NN.
- Each unit will have a **activation function** (a thresholding function) and connections between the units that have weights

Using Sigmoid Unit as an Activation Function



$$o(x) = \sigma(w^T x_i) = \sigma(w_0 + \sum_i w_i x_i) = \frac{1}{1 + \exp(-(w_0 + \sum_i w_i x_i))}$$

Differentiable: $\frac{d\sigma(x)}{dx} = \sigma(x)(1 - \sigma(x))$

Net Activation

$$net_j = \sum_{i=1}^d x_i w_{ji} + w_{j0} = \sum_{i=0}^d x_i w_{ji}$$

where:

- i indexes the inputs units
- j indexes the hidden units
- W_{ji} denotes the weights on input to hidden layer (“synaptic weights”)

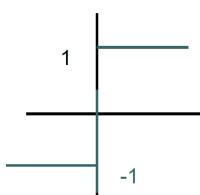
Hidden Layer: Adding Nonlinearity

- Each hidden unit emits an output that is a nonlinear **activation function** of its net activation.

$$y_j = f(\text{net}_j)$$

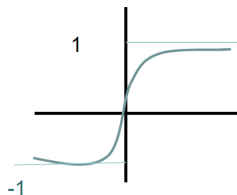
- This is essential to neural networks power, if it's linear then it all becomes just linear regression.
- The output is thus thresholded through this nonlinear activation function.

Activation Functions



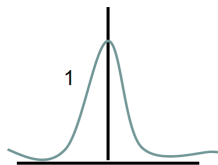
a) Threshold

$$F_T(x) = \begin{cases} 1 & \text{if } x > \tau \\ -1 & \text{otherwise} \end{cases}$$



b) Sigmoid

$$F_S(x) = \frac{1}{(1 + e^{-cx})}$$

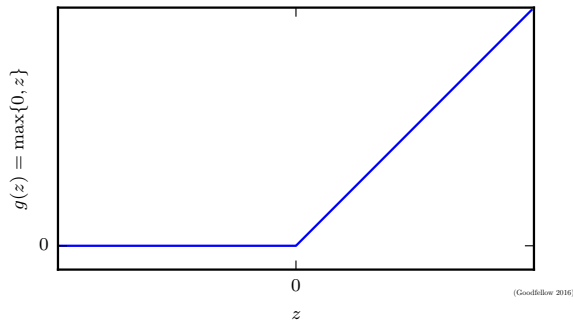


c) Gaussian

$$F_G(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- `tanh` was another common function.
- sigmoid is now discouraged except for final layer to obtain probabilities. Can **over-saturate** easily.
- ReLU is the new standard activation function to use.

Rectified Linear Activation



- **Rectified Linear Units (ReLU)** have become standard $\max(0, net_j)$
 - strong signals are always easy to distinguish
 - most values are zero, derivative is mostly zero
 - they do not saturate as easily as sigmoid
- **new** Exponential linear units - evidence that they perform better than ReLU in some

Output Layer

Regardless of activation function, the output layer then combines all the previous hidden layers.

$$net_k = \sum_{j=1}^{n_H} y_j w_{kj} + w_{k0}$$

where:

- k indexes the output units
- n_H denotes the number of hidden units in the previous layer

Output Layer

The output layer adds *another* level of thresholding, usually nonlinear as well.

$$z_k = f(\text{net}_k)$$

$f(\text{net}_k)$ could be any thresholding function such as the previous activation functions.

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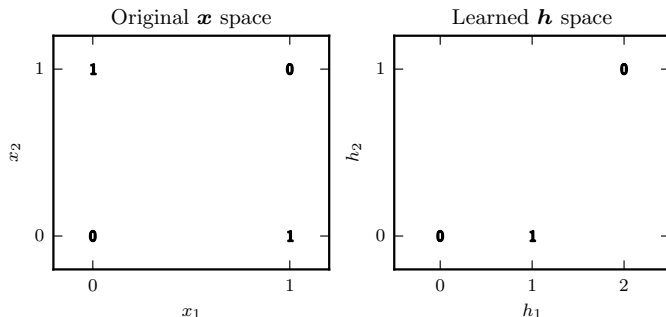
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Failing on the XOR Problem

How is XOR a hard problem? It is if you're too linear. XOR is the simplest **nonlinear** classification problem.

- Two binary features x_1 and x_2 and four patterns to be assigned the correct labels True or False



Neural Nets Can Model XOR Problem

Use a simple `sign` activation function for hidden and output units:

$$f(\text{net}) = 1 \text{ if } \text{net} \geq 0 \\ = -1 \text{ if } \text{net} < 0$$

- First hidden unit:
 - y_1 computes $x_1 + x_2 + 0.5 = 0$
 - $y_1 = 1$ if $\text{net}_1 \geq 0$, $y_1 = -1$ otherwise
- Second hidden unit:
 - y_1 computes $x_1 + x_2 - 1.5 = 0$
 - $y_2 = 1$ if $\text{net}_2 \geq 0$, $y_2 = -1$ otherwise
- single output unit z_1 :
 - emits $z_1 = 1$ iff $y_1 = 1$ and $y_2 = 1$.
 - y_1 models AND gate
 - y_2 models OR gate
 - z_1 models y_1 AND NOT y_2

Full Neural Network Discriminant Function

The class of decision functions that can be described by a three-layer neural network are:

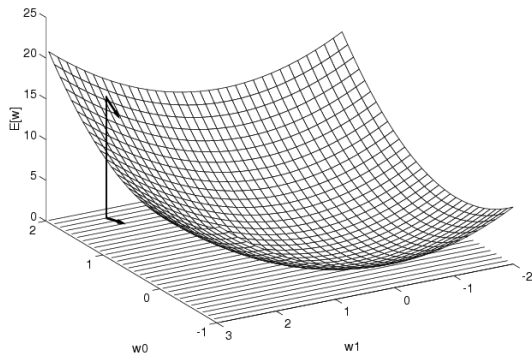
$$g_k(x) = z_k = f \left(\sum_{j=1}^{n_H} w_{kj} f \left(\sum_{i=1}^d w_{ji} x_i + w_{j0} \right) + w_0 \right)$$

- The **Universal Approximation Theorem** [Hornik, 1989] shows that this covers any possible decision function mapping continuous inputs to a finite set of classes to any desired level of accuracy.
- But it does not say how many hidden units would be needed.
- In general, for a single layer it could be exponential (degrees of freedom).
- But, using more layers can reduce the number of hidden units needed and make it more generalizable.

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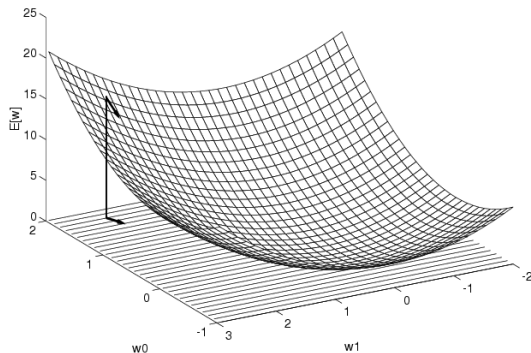
Gradient Descent



Error Function: Mean Squared Error,

cross-entropy loss, etc.

Gradient Descent

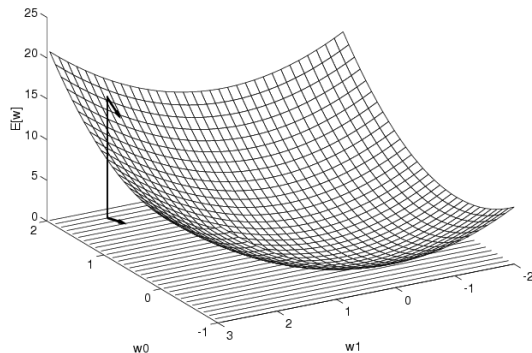


Error Function: Mean Squared Error,

cross-entropy loss, etc.

Gradient: $\nabla E[\mathbf{w}] = \left[\frac{\partial E}{\partial w_0}, \frac{\partial E}{\partial w_1}, \dots, \frac{\partial E}{\partial w_d} \right]$

Gradient Descent



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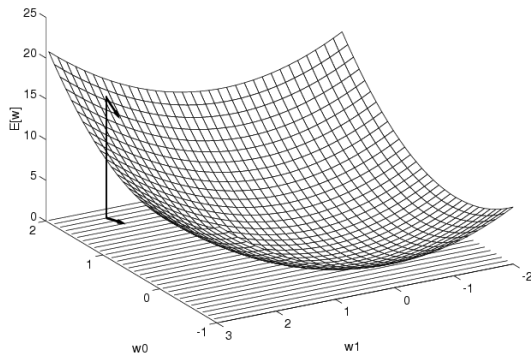
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Training Update Rule: $\Delta w_i = -\eta \frac{\partial E}{\partial w_i}$ where η is the training rate.

Note: For regression, others, this gradient is convex. In NNs it is not. So we must solve

Gradient Descent



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Incremental Gradient Descent

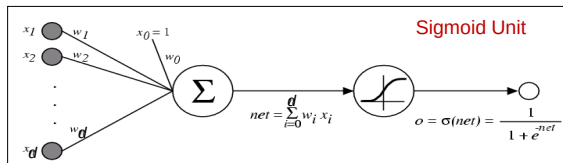
Let error function be : $E_I[\mathbf{w}] = \frac{1}{2}(y^I - o^I)^2$

Do until satisfied:

- For each training example I in D
 - ① Compute the gradient $\nabla E[\mathbf{w}]$
 - ② update weights : $\mathbf{w} = \mathbf{w} - \eta \nabla E[\mathbf{w}]$

Note: can also use batch gradient descent on many points at once.

Error Gradient for a Sigmoid Unit



$$\begin{aligned}
 \frac{\partial E}{\partial w_i} &= \frac{\partial}{\partial w_i} \frac{1}{2} \sum_{l \in D} (y^l - o^l)^2 \\
 &= \frac{1}{2} \sum_l \frac{\partial}{\partial w_i} (y^l - o^l)^2 \\
 &= \frac{1}{2} \sum_l 2(y^l - o^l) \frac{\partial}{\partial w_i} (y^l - o^l) \\
 &= \sum_l (y^l - o^l) \left(-\frac{\partial o^l}{\partial w_i} \right) \\
 &= - \sum_l (y^l - o^l) \frac{\partial o^l}{\partial net^l} \frac{\partial net^l}{\partial w_i}
 \end{aligned}$$

But we know:

$$\begin{aligned}
 \frac{\partial o^l}{\partial net^l} &= \frac{\partial \sigma(net^l)}{\partial net^l} = c^l (1 - c^l) \\
 \frac{\partial net^l}{\partial w_i} &= \frac{\partial (\vec{w} \cdot \vec{x}^l)}{\partial w_i} = x_i^l
 \end{aligned}$$

So:

$$\frac{\partial E}{\partial w_i} = - \sum_{l \in D} (y^l - c^l) o^l (1 - c^l) x_i^l$$

(Slides from Tom Mitchell ML Course, CMU, 2010)

Backpropagation Algorithm

We need an iterative algorithm for getting the gradient efficiently.

For each training example:

- ➊ **Forward propagation**: Input the training example to the network and compute outputs
- ➋ **Compute** output units errors:

$$\delta_k^l = o_k^l(1 - o_k^l)(y_k^l - o_k^l)$$

- ➌ **Compute** hidden units errors:

$$\delta_h^l = o_h^l(1 - o_h^l) \sum_k w_{h,k} \delta_k^l$$

- ➍ **Update** network weights:

$$w_{i,j} = w_{i,j} + \Delta w_{i,j}^l = w_{i,j} + \eta \delta_j^l o_i^l$$

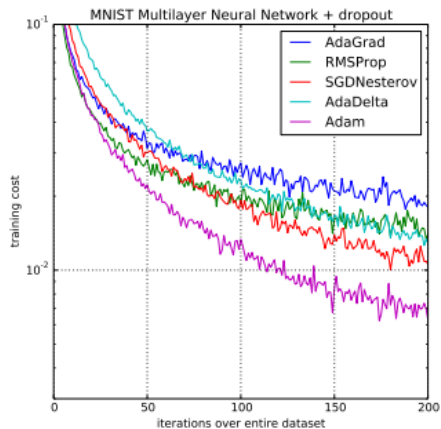
More about Backpropagation

- Still underlies all of Deep Learning
- Can be easily performed on multidimensional outputs.
- No guarantee of global optimal solution, only local, but often good enough.
- Can be parallelized very well, most deep learning tools now do this on GPUs automatically.
- Remember that it is minimizing errors on *training* examples, ability to generalize relies on network structure, training approach, other factors.

Other Optimizers

- **SGD (Stochastic Gradient Descent)**: Can perform best but various additions need to be made and learning rate need to be correctly tuned.
- **Adam (adaptive moment estimate)**: For relatively large datasets Adam is very robust. Converges quickly and gives pretty good performance.
 - keeps track of gradient learning rates and momentum *per dimension*
 - uses variance (2nd moment) rather than means and scales steps based on recent magnitudes
- **L-BFGS** : Converges faster and with better solutions on small datasets. Very memory efficient, scales linearly with variables. Estimates the inverse Hessian matrix (2nd order derivatives)
- See a good overview of the best methods here: [Rud17]
(<https://arxiv.org/abs/1609.04747>)

Adam Optimizer



Lecture Outline

- 1 Recall: Do you Remember Linear Classifiers?
 - Linear Regression
 - Logistic Regression
 - Another Linear Classifier - Naive Bayes
 - Kernel Methods - Turning Linear into Non-linear
- 2 Artificial Neural Networks
 - Simple Neural Network Definition
- 3 History
- 4 How Simple Neural Networks Work
- 5 Neural Networks as Universal Approximators
- 6 Training Neural Networks - Backpropagation
- 7 References and Further Reading

References



Sebastian Ruder, *An overview of gradient descent optimization algorithms*, [arXiv:1609.04747 \[cs\]](#) (2017), [arXiv: 1609.04747](#).